Ranker: Fast Variational Bayesian Ranking from Pairwise Comparisons

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Abstract

Ranker is a Bayesian algorithm for ranking items in a set, inspired by the ELO rating system in chess. It assumes each item has a latent score, normally distributed with a latent variance, and that the outcome of a pairwise comparison is a Bernoulli trial with a probability of success given by the logistic function of the difference in scores.

Ranker infers the latent scores of the items and the variance using a variational Bayes approximation which models the posterior as an inverse-gamma distributed variance and independent normally distributed scores. We compute the KL-divergence, its gradient and Hessian in closed form by approximating the logistic function with the error function. This yields a very fast inference technique, based on a damped Newton conjugate-gradient iteration.

Ranker can do more than infer scores and rank items, it can also be used to select the most useful pair of items to compare next. We do this efficiently by differentiating the fitted posterior with respect to the observed trials through the implicit function theorem. A single inference, and a single round of next-pair selection, both run in $\mathcal{O}(n^2)$ time, where n is the number of items. We validate the approximation against a reference Markov-chain Monte Carlo posterior.

1 Introduction

Ranking a set of items from pairwise comparisons is a recurring problem: rating chess players from game outcomes, ordering products from A/B preferences, or eliciting a person's taste by repeatedly asking them "which of these two do you prefer?". Two questions arise. First, given a collection of noisy comparisons, what is the best estimate of the underlying ranking, and how confident are we in it? Second — and this is what makes the interactive setting interesting — which comparison should we ask for next so as to learn the most?

Ranker answers both. It is a Bayesian model in which each item carries a latent score, scores share a latent variance drawn from an inverse-gamma prior, and the outcome of a comparison is a Bernoulli trial whose success probability is the logistic function of the score difference. This is the Bradley–Terry model [BT52] with a hierarchical Gaussian prior on the scores.

Exact inference in this model is intractable, so we use a mean-field variational Bayes approximation [BKM17]: the posterior is approximated by an inverse-gamma distribution over the variance and a product of independent Gaussians over the scores. The contribution of this note is that, by replacing the logistic link with a closely matched error-function surrogate, *every* term of the KL divergence — including its gradient and Hessian — can be written in closed form using elementary and polygamma functions. The resulting objective is minimized with a damped Newton conjugate-gradient iteration whose Hessian-vector products cost $\mathcal{O}(n + \mathrm{nnz}(o))$ — linear in the number of items and observed pairs — thanks to the sparsity structure of the problem.

The same closed form makes active learning cheap. Because the fitted parameters are a smooth function of the observation counts, we differentiate them with respect to a hypothetical comparison through the implicit function theorem and pick the pair whose outcome is expected to most improve a chosen criterion. This is a form of Bayesian optimal experimental design, specialised to pairwise comparisons; empirically, uncertainty-directed criteria (information gain or posterior variance) select comparisons more efficiently than greedily minimising the ranking loss itself, which is myopic.

Our contributions are:

- a closed-form mean-field variational objective for hierarchical Bradley–Terry, obtained through an error-function approximation of the logistic-normal integral [Cro09] (Sections 5);

- the sparse gradient/Hessian structure that makes one inference cost $\mathcal{O}(n^2)$ (Section 5.5);

- a differentiable next-pair selection rule for active comparison (Section 7); and

- an empirical validation against an exact-model MCMC reference, and a comparison of acquisition criteria for active pair selection (Section 8).

2 Related work

The Bradley–Terry model [BT52] is the canonical model of paired comparisons; [Hun04] gives efficient MM algorithms for its maximum likelihood estimation and its generalisations. The ELO system [Elo78] is an online point-update heuristic for the same likelihood, and Glicko [Gli99] augments it with a rating-reliability term, taking a step towards a Bayesian treatment. BayesElo applies Maximum a Posteriori estimation on top of the Bradley–Terry likelihood.

The closest relative of Ranker is TrueSkill [HMG07], a Bayesian skill rating system that maintains a Gaussian belief over each player's skill and updates it by approximate message passing (expectation propagation) on a factor graph. Ranker differs in three respects: it places a shared hierarchical prior with an *inferred* variance over the scores rather than fixing a per-player prior; it uses a global mean-field variational objective minimized to convergence rather than a single online pass; and it exposes the fitted posterior as a differentiable function of the data, which is what enables the active next-pair selection. The error-function approximation to the logistic-normal integral we rely on is due to Crooks [Cro09].

2.1 Comparison with Pólya–Gamma augmentation

The standard route to Bayesian inference in logistic and Bradley–Terry models is Pólya–Gamma data augmentation [PSW13]: one introduces an auxiliary Pólya–Gamma variable for each compared pair, conditional on which the logistic likelihood becomes Gaussian in the scores. The resulting conjugacy gives simple closed-form updates and underlies both Gibbs samplers and mean-field variational schemes for these models. Ranker takes a different route — replacing the logistic link by an error-function surrogate rather than augmenting it — with three practical consequences.

First, there are no augmentation variables to infer. Pólya–Gamma carries one latent variable per compared pair, which must be resampled (Gibbs) or mean-field averaged (variational) at every sweep, and whose mean-field treatment is an extra factorisation layered on top of the one over the scores.

The surrogate removes it entirely: the observations enter only through the fixed count matrix o, and the variational family is over the scores and the shared variance alone, giving the low-dimensional closed-form objective of Section 5.

Second, that objective, its gradient, and its Hessian are all closed form, so it is minimised with a damped Newton conjugate-gradient iteration — converging in a handful of second-order steps — rather than the first-order coordinate updates that augmentation-based variational inference relies on, which mix slowly under the coupling of the hierarchical prior.

Third, the augmentation-free objective makes the fitted posterior a smooth, differentiable function of the observation counts, which is exactly what enables the active next-pair selection of Section 7 by differentiating through the fit. Under a Pólya–Gamma treatment the objective is defined only after the auxiliary variables are integrated or averaged out, so differentiating the fit with respect to the data is far less direct.

The price is a controlled approximation: a Pólya–Gamma Gibbs sampler is asymptotically exact, whereas the error-function surrogate is not. As Section 5.3 shows, however, its error can be made arbitrarily small with a fast-converging sequence of corrections, at which point the mean-field factorisation — which any variational scheme, Pólya–Gamma included, must also make — is the dominant approximation, not the link.

3 Background

Ranker is inspired by the ELO rating system in chess [Elo78]. In chess, each player has a rating which is a positive integer. The rating of a player is a measure of their skill, and it is updated after each game. Each player's rating is updated according to the following formula:

$$R_{new} = R_{old} + K(S - E) \tag{1}$$

where R_{new} and R_{old} are the new and old ratings, K is a constant which determines how much the rating changes, S is the score of the player in the game, and E is the expected score of the player, given the ratings of the two players. The expected score is computed as follows:

$$E = \frac{1}{1 + 10^{-\frac{R_{other} - R_{self}}{400}}} \tag{2}$$

where R_{other} is the rating of the opponent, and R_{self} is the rating of the player.

This is an approximate update rule which is only valid when ratings are close to each other. It corresponds to a gradient descent algorithm with a learning rate of K. The lack of decay in the learning rate is a two-edged sword: it allows the ratings to change quickly, reflecting a player's recent performance, but it also means that the ratings can change wildly. For that reason, K is typically set at 16 for masters and 32 for grandmasters.

A partial Bayesian approach to the ELO rating system is described in BayesElo, using Maximum a Posteriori (MAP) estimation, following the work of [Hun04].

4 Approach

4.1 Model

Ranker assumes that each item has a latent score, normally distributed with a latent variance. The outcome of a pairwise comparison is a Bernoulli trial with a probability of success given by the logistic function of the difference. We write $\mathcal{N}(\mu, \sigma^2)$ for the normal distribution of mean μ and *variance* σ^2. The model is as follows:

$$\begin{aligned} \nu &\sim \text{InvGamma}(\alpha_h, \beta_h) \\ \forall i, z_i &\sim \mathcal{N}(0, \nu) \\ \forall i < j, o_{i,j} &\sim \text{Binomial}\left(\frac{1}{1 + e^{-(z_i - z_j)}}, m_{i,j}\right) \\ \forall i > j, o_{i,j} &= m_{i,j} - o_{j,i} \end{aligned} \tag{3}$$

where α_h and β_h are hyperparameters, and $o_{i,j}$ is the number of wins of item i over item j out of $m_{i,j}$ comparisons between items i and j, where $m_{i,j} = m_{j,i}$ is fixed and given. Note that here ν is the *variance* of the scores. We use the convention that β_h represents a rate, not a scale: the inverse-gamma density is $p(\nu) = \frac{\beta_h^{\alpha_h}}{\Gamma(\alpha_h)} \nu^{-\alpha_h - 1} e^{-\beta_h/\nu}$, so the scale is $1/\beta_h$.

4.1.1 Choice of hyperparameters

We pick concrete values for the hyperparameters $\alpha_h = 1.2$ and $\beta_h = 2$ (equivalently, scale $1/\beta_h = 0.5$). A hand-wavy justification for these values follows.

Consider the binomial distribution over 10 stars, obtained by flipping a coin ten times and summing the number of heads. It has a standard deviation of $\sqrt{5/2} \simeq 1.5811$. The difference between two adjacent star ratings is thus $\sqrt{2/5} \simeq 0.6325$. Based on this intuition, we call a difference of $\sqrt{2/5}$ standard deviations a "star" of difference.

Assuming this is an intuitive notion of a "star", what odds should an additional star confer to an item in a head-to-head match? If those comparisons are hard to judge and very subjective, maybe only 55% of the time? If they're very clear, perhaps over 99% of the time?

Using a logistic rule:

- 55% would correspond to a standard deviation of $\sqrt{5/2}\log(11/9) \simeq 0.317$
- 99% would correspond to a standard deviation of $\sqrt{5/2}\log(19) \simeq 4.656$

None of this is very rigorous, but it gives us some sense of the scale of standard deviations we're dealing with. Clearly 0.0001 and 100 are silly.

These roughly correspond to the 1% and 99% tails of the $\mathrm{InvGamma}(1.2, 2)$ distribution for the variance, and justify our choice of hyperparameters. We are proud to incorporate actual prior knowledge, as opposed to copping out of it by using a so-called "uninformative" prior :-)

4.2 Variational Bayes approximation

We approximate the posterior with a factorized (mean-field) distribution Q,

$$\begin{aligned} \nu &\sim \mathrm{InvGamma}(\alpha, \beta) \\ \forall i, z_i &\sim \mathcal{N}(\mu_i, \sigma_i^2) \end{aligned} \tag{4}$$

so that σ_i is the standard deviation of the marginal over z_i. Note that α and β are parameters of the approximation to the posterior, not the hyperparameters of the model, which are α_h and β_h. Nonetheless, we naturally initialize α and β to α_h and β_h. We initialize μ_i to 0. The case of σ_i is more delicate. Absent any observation, the marginal distribution of z_i is a Student's t distribution with $2\alpha_h$ degrees of freedom and scale $\sqrt{\beta_h/\alpha_h}$. We choose to initialize σ_i to minimize the KL divergence between the Gaussian marginal and that Student distribution. We discovered empirically (but did not attempt to prove) that the minimum is reached, to leading order in $1/\alpha_h$, for

$$\sigma_i = \sqrt{\frac{\beta_h}{\alpha_h}} \left(1 + \frac{3}{6\alpha_h + 2\alpha_h^2} + \mathcal{O}(\alpha_h^{-3}) \right).$$

5 Closed-form expression

5.1 KL-divergence

The KL divergence between the true posterior, P, and the variational approximation Q is $D_{KL}(Q\|P) = -H(Q) + H(Q,P)$, where $H(Q)$ is the entropy of Q and $H(Q,P)$ is the cross entropy of Q and P. The entropy of Q is

$$\begin{aligned} H(Q) = &\int_{\nu=0}^{\infty} \frac{\beta^{-\alpha}\nu^{-\alpha-1}e^{-\frac{1}{\beta\nu}}}{\Gamma(\alpha)} \\ &\times \left(\alpha \log(\beta) + (\alpha+1)\log(\nu) + \frac{1}{\beta\nu} + \log(\Gamma(\alpha)) \right) d\nu \\ &+ \sum_{i=1}^{n} \int_{z_i=-\infty}^{\infty} \frac{e^{-\frac{1}{2}\left(\frac{z_i-\mu_i}{\sigma_i}\right)^2}}{\sigma_i\sqrt{2\pi}} \\ &\times \left(\frac{1}{2}\log 2\pi + \log\sigma_i + \frac{1}{2}\left(\frac{z_i-\mu_i}{\sigma_i}\right)^2 \right) dz_i \end{aligned}$$

The cross entropy of Q and P is

$$\begin{aligned} H(Q,P) = &\int_{\nu=0}^{\infty} \frac{\beta^{-\alpha}\nu^{-\alpha-1}e^{-\frac{1}{\beta\nu}}}{\Gamma(\alpha)} \\ &\times \left(\alpha_h \log(\beta_h) + (\alpha_h+1)\log(\nu) + \frac{1}{\beta_h\nu} + \log(\Gamma(\alpha_h)) \right) d\nu \\ &+ \sum_{i=1}^{n} \int_{\nu=0}^{\infty} \int_{z_i=-\infty}^{\infty} \frac{e^{-\frac{1}{2}\left(\frac{z_i-\mu_i}{\sigma_i}\right)^2}}{\sigma_i\sqrt{2\pi}} \frac{\beta^{-\alpha}\nu^{-\alpha-1}e^{-\frac{1}{\beta\nu}}}{\Gamma(\alpha)} \\ &\times \left(\frac{1}{2}\log 2\pi + \frac{1}{2}\log(\nu) + \frac{1}{2}\frac{z_i^2}{\nu} \right) dz_i d\nu \end{aligned}$$

$$+\sum_{i=1}^{n}\sum_{j=1}^{n} o_{i,j} \int_{z_\delta=-\infty}^{\infty} \frac{e^{-\frac{1}{2}\left(\frac{z_\delta-(\mu_i-\mu_j)}{\sqrt{\sigma_i^2+\sigma_j^2}}\right)^2}}{\sqrt{\sigma_i^2+\sigma_j^2}\sqrt{2\pi}} \times \log\left(1+e^{-z_\delta}\right) dz_\delta + Cst$$

The constant Cst collects the normalising terms that do not depend on the variational parameters; it does not affect the minimiser.

5.2 The logistic-normal integral

As it turns out, we can express most of the integrals above in terms of elementary functions, as well as the polygamma function ψ. The integral

$$\int_{z_\delta=-\infty}^{\infty} \frac{e^{-\frac{1}{2}\left(\frac{z_\delta-(\mu_i-\mu_j)}{\sqrt{\sigma_i^2+\sigma_j^2}}\right)^2}}{\sqrt{\sigma_i^2+\sigma_j^2}\sqrt{2\pi}} \log\left(1+e^{-z_\delta}\right) dz_\delta$$

is more problematic. A related integral is the logistic-normal integral,

$$\int_{z_\delta=-\infty}^{\infty} \frac{e^{-\frac{1}{2}\left(\frac{z_\delta-(\mu_i-\mu_j)}{\sqrt{\sigma_i^2+\sigma_j^2}}\right)^2}}{\sqrt{\sigma_i^2+\sigma_j^2}\sqrt{2\pi}} \frac{1}{1+e^{-z_\delta}} dz_\delta.$$

[Cro09] offers an approximation by replacing the logistic function with an error function:

$$\frac{1}{1+e^{-z}} \simeq \frac{1}{2} + \frac{1}{2}\mathrm{erf}\left(\frac{\sqrt{\pi}}{4}z\right).$$

Given that

$$\frac{d}{dz}\log\left(1+e^{-z}\right) = \frac{1}{1+e^{-z}} - 1,$$

we choose to approximate $\log\left(1+e^{-z}\right)$ as

$$\log\left(1+e^{-z}\right) \simeq \int_{\zeta=z}^{\infty} \left(\frac{1}{2} - \frac{1}{2}\mathrm{erf}\left(\frac{\sqrt{\pi}}{4}\zeta\right)\right) d\zeta = \frac{2}{\pi}e^{-\frac{\pi z^2}{16}} - \frac{1}{2}z\ \mathrm{erfc}\left(\frac{\sqrt{\pi}}{4}z\right).$$

The difference between the two is maximal for $z = 0$ and is $\log(2) - \frac{2}{\pi} \simeq 0.0565$. The difference between the derivatives with respect to z is at most 0.0177. Figure 1 shows both the function and the error.

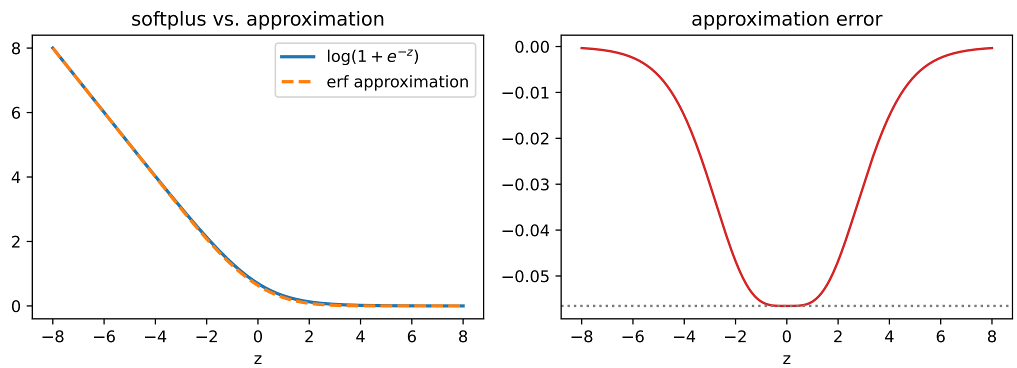

Figure 1: Left: the softplus $\log(1+e^{-z})$ and its error-function approximation. Right: the approximation error, which is bounded by $\log(2) - \frac{2}{\pi} \simeq 0.0565$ (dotted line) and attained at $z = 0$.

5.3 Refining the approximation

The error-function approximation is convenient, but an error of $\simeq 0.0565$ is not negligible. We can drive it down by orders of magnitude essentially for free, while keeping every integral in closed form. The error $r(z) = \log(1 + e^{-z}) - \widehat{f}(z)$ of the approximation $\widehat{f}$ is an even, non-negative function that peaks at $z = 0$ and decays. We cancel it with a sum of k Gaussian bumps,

$$\varepsilon(z) = \sum_{i=1}^{k} c_i \left(1 + a_i z^2\right) e^{-a_i z^2},$$

each of which *still* integrates against a Gaussian in closed form: for the model's posterior over the score difference, $\mathcal{N}(\mu_\delta, \sigma_\delta^2)$, writing $D_i = 1+2a_i\sigma_\delta^2$,

$$\int \mathcal{N}(z; \mu_\delta, \sigma_\delta^2)\, c_i \left(1 + a_i z^2\right) e^{-a_i z^2}\, dz = \frac{c_i}{\sqrt{D_i}}\, e^{-a_i \mu_\delta^2 / D_i} \left(1 + \frac{a_i \mu_\delta^2}{D_i^2} + \frac{a_i \sigma_\delta^2}{D_i}\right).$$

So the corrected observation term, its gradient, and its Hessian remain elementary closed-form expressions (no new special functions, and the same

$D + R + S$ sparsity of Section 5.5); the correction is simply added to the per-observation kernels.

We fit the coefficients $\{a_i, c_i\}$ to minimize the worst-case (minimax) absolute error of $\widehat{f} + \varepsilon$. With the widths a_i fixed the c_i enter linearly, so the minimax fit of c_i is a linear program; we search the k widths globally and then refine (a_i, c_i) jointly by gradient descent on a smooth L_p surrogate of the (non-smooth) maximum. The fits are also monotone improvements — the corrected error is no larger than the original at *every* z. Table 1 lists the coefficients and worst error for $k = 1, \ldots, 7$; each extra term cuts the worst error several-fold (about $10\times$ for the first terms, tapering to $\sim 4\times$ by $k = 7$). Figure 2 shows the $k = 5$ case, where the error falls from 0.0565 to $\simeq 3 \times 10^{-7}$ — five orders of magnitude — at which point the error-function approximation is no longer a meaningful source of error next to the mean-field variational approximation itself.

k	worst error	a_i	c_i
1	1.555×10^{-3}	1.618×10^{-1}	5.497×10^{-2}
2	1.233×10^{-4}	$1.028\times10^{-1}, 2.256\times10^{-1}$	$2.052\times10^{-2}, 3.588\times10^{-2}$
3	1.331×10^{-5}	$7.432\times10^{-2}, 1.589\times10^{-1}, 2.970\times10^{-1}$	$6.511\times10^{-3}, 3.585\times10^{-2}, 1.415\times10^{-2}$
4	1.695×10^{-6}	$5.812\times10^{-2}, 1.157\times10^{-1}, 2.039\times10^{-1}, 3.774\times10^{-1}$	$2.110\times10^{-3}, 1.832\times10^{-2}, 3.142\times10^{-2}, 4.673\times10^{-3}$
5	2.861×10^{-7}	$4.802\times10^{-2}, 8.967\times10^{-2}, 1.575\times10^{-1}, 2.527\times10^{-1}, 4.631\times10^{-1}$	$7.260\times10^{-4}, 8.549\times10^{-3}, 2.780\times10^{-2}, 1.792\times10^{-2}, 1.535\times10^{-3}$
6	7.232×10^{-8}	$3.846\times10^{-2}, 6.758\times10^{-2}, 1.143\times10^{-1}, 1.846\times10^{-1}, 2.919\times10^{-1}, 5.191\times10^{-1}$	$1.672\times10^{-4}, 2.907\times10^{-3}, 1.415\times10^{-2}, 2.836\times10^{-2}, 1.020\times10^{-2}, 7.423\times10^{-4}$
7	1.887×10^{-8}	$3.542\times10^{-2}, 6.007\times10^{-2}, 9.706\times10^{-2}, 1.527\times10^{-1}, 2.257\times10^{-1}, 3.568\times10^{-1}, 6.050\times10^{-1}$	$8.874\times10^{-5}, 1.631\times10^{-3}, 8.478\times10^{-3}, 2.184\times10^{-2}, 2.012\times10^{-2}, 4.125\times10^{-3}, 2.470\times10^{-4}$

Table 1: Correction coefficients fit by minimax, for $k = 1, \ldots, 7$ terms. "Worst error" is the maximum absolute error of $\widehat{f} + \varepsilon$ over z. Each added term reduces it several-fold ($\sim 10\times$ early, tapering to $\sim 4\times$ by $k = 7$). In practice we use $k = 5$.

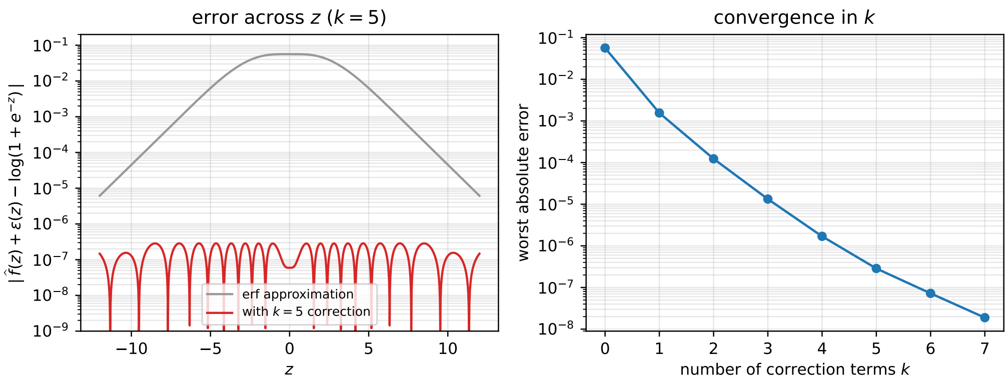

Figure 2: Left: absolute error $|\widehat{f}(z)+\varepsilon(z)-\log(1+e^{-z})|$ of the error-function approximation alone (grey) and after the five-term correction (red), on a logarithmic scale; the dips in the red curve are the zero-crossings of the equioscillating minimax error. Right: the worst absolute error over z against the number of correction terms k ($k = 0$ is the plain error-function approximation), showing the roughly geometric convergence — each term shrinks the error several-fold.

5.4 Putting it all together

After carrying out the integrals, the part of the divergence that does not depend on the observations is

$$\underbrace{\frac{\alpha\beta}{\beta_h} - \alpha + \alpha_h \log\frac{\beta_h}{\beta} + \log\Gamma(\alpha_h) - \log\Gamma(\alpha) + (\alpha - \alpha_h)\,\psi(\alpha)}_{\text{inverse-gamma KL}}$$
$$+ \underbrace{\frac{1}{2}\,\alpha\beta\sum_{i=0}^{n-1}\left(\mu_i^2 + \sigma_i^2\right) - \sum_{i=0}^{n-1}\log\sigma_i - \frac{n}{2}\big(1 + \log(\beta) + \psi(\alpha)\big)}_{\text{Gaussian KL}}.$$

The $\frac{1}{2}\log 2\pi$ constants of the Gaussian entropy and cross-entropy cancel exactly, which is why no $\log 2\pi$ term remains.

Then, to account for all observed trials, we add

$$\sum_{i=0}^{n-1}\sum_{j=0}^{n-1} o_{i,j}\cdot\left(e^{-\frac{\mu_\delta^2}{h^2}}\frac{h}{2\sqrt{\pi}}+\mu_\delta g_{\mu_\delta}+\mathbb{E}_{z\sim\mathcal{N}(\mu_\delta,\sigma_\delta^2)}[\varepsilon(z)]\right)$$

where

$$\begin{aligned}
\mu_\delta &= \mu_i-\mu_j,\\
\sigma_\delta &= \sqrt{\sigma_i^2+\sigma_j^2},\\
h &= \sqrt{\frac{16}{\pi}+2\sigma_\delta^2},\\
g_{\mu_\delta} &= -\frac{1}{2}\left(1-\operatorname{erf}\left(\frac{\mu_\delta}{h}\right)\right).
\end{aligned}$$

The contribution of each observed pair is $o_{i,j}\,\mathbb{E}[\log(1+e^{-z})]$, where the expectation runs over that pair's posterior $z\sim\mathcal{N}(\mu_\delta,\sigma_\delta^2)$; all expectations below are over it. The first two terms in the summand are the closed form of this under the error-function approximation, and the third is the same expectation of the correction $\varepsilon(z)$ of Section 5.3. Note that $\varepsilon(z)$ is a function of z alone: its dependence on the pair, and any coupling between μ_δ and σ_δ, enters only through the posterior it is averaged against, exactly as for the error-function term. In closed form, with $D_\ell = 1+2a_\ell\sigma_\delta^2$,

$$\begin{aligned}
\mathbb{E}[\varepsilon(z)] &= \int \mathcal{N}(z;\mu_\delta,\sigma_\delta^2)\,\varepsilon(z)\,dz\\
&= \sum_{\ell=1}^{k}\frac{c_\ell}{\sqrt{D_\ell}}\,e^{-a_\ell\mu_\delta^2/D_\ell}\left(1+\frac{a_\ell\mu_\delta^2}{D_\ell^2}+\frac{a_\ell\sigma_\delta^2}{D_\ell}\right).
\end{aligned}$$

Dropping it recovers the plain error-function objective.

Note that $o_{i,j}$ is always positive: it represents the number of times i beat j. This is not a signed quantity. This is indeed important; i beating j and then j beating i over the course of two trials is not the same as no trial at all.

5.5 Gradient and Hessian

The gradient and Hessian of the objective with respect to the variational parameters $\theta=(\mu_1,\ldots,\mu_n,\sigma_1,\ldots,\sigma_n,\alpha,\beta)$ follow by direct differentiation of the closed form above. Before writing them out it is worth describing their *structure*, because that is what makes inference fast. Both split into an observation-independent part and a sum of one local contribution per observed pair, and the Hessian decomposes as

$$H = D + R + S,$$

where:

- D is diagonal (the observation-independent terms together with the diagonal of the observation terms);
- R is a rank-coupling block confined to the two scale parameters (α, β) and their interaction with each (μ_i, σ_i) — i.e. dense only in the last two rows and columns, and of rank at most two there;
- S is sparse, with a constant number of off-diagonal entries ((μ_i, μ_j), (σ_i, σ_j), (μ_i, σ_j), (μ_j, σ_i), $\ldots$) per observed pair $o_{i,j} > 0$.

Consequently a Hessian-vector product Hx costs $\mathcal{O}(n+\mathrm{nnz}(o))$, where $\mathrm{nnz}(o)$ is the number of observed pairs. We never form H explicitly; we expose it as a linear operator and solve $Hx = g$ with conjugate gradients (Section 6).

Observation-independent terms. Writing $\psi_1 = \psi'$ for the trigamma function and $\Sigma = \sum_i (\mu_i^2 + \sigma_i^2)$, the gradient of the observation-free part is

$$\partial_{\mu_i} = \alpha\beta\,\mu_i, \qquad \partial_{\sigma_i} = \alpha\beta\,\sigma_i - \frac{1}{\sigma_i},$$
$$\partial_\alpha = \frac{\beta}{\beta_h} - 1 + \tfrac{1}{2}\beta\,\Sigma + \left(\alpha - \alpha_h - \tfrac{n}{2}\right)\psi_1(\alpha), \qquad \partial_\beta = \frac{\alpha}{\beta_h} + \tfrac{1}{2}\alpha\,\Sigma - \frac{\alpha_h + \frac{n}{2}}{\beta}.$$

Its Hessian is diagonal in the (μ, σ) block, $\partial^2_{\mu_i} = \alpha\beta$ and $\partial^2_{\sigma_i} = \alpha\beta + 1/\sigma_i^2$, with the only couplings being the (α, β) corner and the rank-two coupling of (α, β) to each μ_i, σ_i (namely $\partial_{\mu_i}\partial_\alpha = \beta\mu_i$, $\partial_{\mu_i}\partial_\beta = \alpha\mu_i$, and likewise for σ_i); these are the blocks D and R.

Per-observation terms. Each observed pair contributes $o_{i,j}\,F(\mu_\delta, \sigma_\delta)$ to the objective, with $\mu_\delta = \mu_i - \mu_j$, $\sigma_\delta = \sqrt{\sigma_i^2 + \sigma_j^2}$, $h = \sqrt{16/\pi + 2\sigma_\delta^2}$, and

$F = \frac{h}{2\sqrt{\pi}} e^{-\mu_\delta^2/h^2} - \frac{1}{2}\mu_\delta\big(1 - \operatorname{erf}(\mu_\delta/h)\big)$. Its first derivatives in the two natural variables are

$$F_{\mu_\delta} = -\tfrac{1}{2}\big(1 - \operatorname{erf}(\mu_\delta/h)\big), \qquad F_{\sigma_\delta} = \frac{\sigma_\delta}{\sqrt{\pi}\,h}\, e^{-\mu_\delta^2/h^2},$$

and the second derivatives are

$$F_{\mu_\delta\mu_\delta} = \frac{e^{-\mu_\delta^2/h^2}}{\sqrt{\pi}\,h}, \qquad F_{\mu_\delta\sigma_\delta} = -\frac{2\,\mu_\delta\,\sigma_\delta\,e^{-\mu_\delta^2/h^2}}{\sqrt{\pi}\,h^3},$$
$$F_{\sigma_\delta\sigma_\delta} = \frac{\big(256 + 4\pi\sigma_\delta^2(8 + \pi\mu_\delta^2)\big)e^{-\mu_\delta^2/h^2}}{\pi^{5/2}\,h^5}.$$

When the refined approximation of Section 5.3 is used, each of these kernels simply gains the corresponding derivative of the correction term — also a closed-form, elementary expression of $(\mu_\delta, \sigma_\delta)$ — with no change to the structure. The entries of S (and the observation part of D) then follow by the chain rule through μ_δ and $\sigma_\delta = \sqrt{\sigma_i^2 + \sigma_j^2}$, using $\partial\sigma_\delta/\partial\sigma_i = \sigma_i/\sigma_\delta$, $\partial^2\sigma_\delta/\partial\sigma_i^2 = \sigma_j^2/\sigma_\delta^3$, and $\partial^2\sigma_\delta/\partial\sigma_i\partial\sigma_j = -\sigma_i\sigma_j/\sigma_\delta^3$; for example $\partial_{\mu_i} = -\partial_{\mu_j} = o_{i,j}F_{\mu_\delta}$, while $\partial^2_{\mu_i} = o_{i,j}F_{\mu_\delta\mu_\delta}$ and $\partial_{\mu_i}\partial_{\mu_j} = -o_{i,j}F_{\mu_\delta\mu_\delta}$.

6 Inference

We minimize the KL divergence with a damped Newton conjugate-gradient iteration. At each step we solve the Newton system $Hx = g$ approximately with conjugate gradients, using the Hessian-vector product of Section 5.5, and we precondition with the (Jacobi) diagonal of H. The positive parameters σ_i, α, β are updated in the log domain so that they remain positive. As in the Levenberg–Marquardt method, we add a damping term λ to the diagonal and increase λ geometrically whenever a step fails to decrease the objective, decreasing it again once steps succeed. This combines the fast local convergence of Newton's method with the robustness of gradient descent far from the optimum.

Why conjugate gradients fit this problem so well. The Hessian is exactly of the form a preconditioned conjugate-gradient solver likes best. After Jacobi (diagonal) preconditioning, the observation-free part $D + R$ becomes the identity plus a rank-two coupling to (α, β), and the observation part

S adds off-diagonal entries along the (typically sparse) comparison graph. The result is a well-conditioned operator whose spectrum is tightly clustered around one, with only a handful of outlying eigenvalues from the low-rank coupling and the comparison structure. Conjugate gradients converges in a number of iterations governed by the number of eigenvalue *clusters*, not by the dimension, so the iteration count stays essentially constant as n grows: empirically each Newton solve takes about 10–12 conjugate-gradient iterations whether $n = 10$ or $n = 200$, and the damped-Newton loop itself converges in fewer than ten steps. Because the method is matrix-free — it needs only the $\mathcal{O}(n + \mathrm{nnz}(o))$ Hessian-vector product and never a factorization — this turns each inference into a small constant number of cheap structured matrix-vector products. The same property is what makes active selection cheap: it needs H^{-1} applied to a single vector (the loss gradient), which is exactly what one conjugate-gradient solve delivers, and that solve is then reused across all candidate pairs (Section 7).

7 Active pair selection

7.1 Acquisition criteria

Having a posterior, we can ask which comparison to request next. We consider three criteria computed from the fitted posterior:

- the *expected number of inversions*, i.e. the expected number of pairs whose posterior mean order disagrees with their true order. For a pair (i, j) this is $\frac{1}{2}\,\mathrm{erfc}\big(|\mu_i - \mu_j|/(\sqrt{2}\,\sigma_\delta)\big)$, and the total is the sum over all pairs. This is the task loss itself, and it is differentiable in (μ, σ); a candidate pair is scored by its expected reduction.

- the *sum of posterior variances* $\sum_i \sigma_i^2$, a simple uncertainty proxy, likewise differentiable and scored by expected reduction.

- the *expected information gain* of the next comparison (the BALD criterion): the mutual information between its binary outcome y and the latent scores, $\mathrm{I}(y; z) = \mathrm{H}[\bar{p}_{ij}] - \mathbb{E}[\mathrm{H}[\sigma(z_i - z_j)]]$, where $\mathrm{H}[p] = -p \log p - (1-p)\log(1-p)$ is the binary entropy, $\bar{p}_{ij} = \mathbb{E}[\sigma(z_i - z_j)]$ is the posterior-averaged win probability, and the expectation is over the one-dimensional Gaussian posterior of $z_i - z_j$, evaluated by Gauss–Hermite quadrature. It is *maximised*, and — unlike the two losses — it is read off each pair directly rather than through the fit.

7.2 Differentiating through the fit

At a minimiser the gradient of the objective vanishes, $g(\theta^\star, o) = 0$. By the implicit function theorem, the fitted parameters depend smoothly on the observation counts o, with

$$\frac{\partial \theta^\star}{\partial o_{i,j}} = -H^{-1} \frac{\partial g}{\partial o_{i,j}},$$

where H is the same Hessian used for inference and $\partial g / \partial o_{i,j}$ is cheap because the objective is linear in o. Given a loss $f(\theta)$, the marginal effect of a hypothetical observation is then obtained by the chain rule,

$$\frac{\partial f}{\partial o_{i,j}} = \left(\frac{\partial f}{\partial \theta}\right)^\top \frac{\partial \theta^\star}{\partial o_{i,j}} = -\left(\frac{\partial f}{\partial \theta}\right)^\top H^{-1} \frac{\partial g}{\partial o_{i,j}}.$$

A single conjugate-gradient solve of $H^{-1} \partial f / \partial \theta$ is shared across all candidate pairs, after which each pair is scored in $\mathcal{O}(1)$. We weight the win and loss outcomes of the candidate comparison by their posterior probabilities (computed in closed form from the approximation) to obtain the expected gain, and pick the pair with the largest expected gain.

The same gradient can be used as a heuristic to guide a deeper tree search over sequences of future comparisons, and a candidate pair's predicted gain can be verified by actually refitting the model with the hypothetical observation added, at the cost of one extra fit per pair.

8 Experiments

We validate the implementation on synthetic data drawn from the model.

Posterior accuracy. To check the combined effect of the mean-field factorisation and the error-function approximation, we compare the variational posterior to a reference obtained by random-walk Metropolis sampling of the *exact* logistic posterior (no approximation). For $n = 12$ items with a dense set of comparisons, Figure 3 shows that the variational posterior means track the MCMC means almost exactly, and that the per-item standard deviations agree once scores are compared on the identified (relative) scale. The one thing mean-field VB cannot represent is the correlation between scores — in particular the shared, near-global shift that the weak hierarchical prior leaves loosely constrained — but that shift is common to every item and irrelevant to the ranking, so the identified marginals VB reports match the reference. We do not compare the scale ν itself: at $n = 12$ it is only weakly identified, and both methods place broad, data-driven posteriors on it.

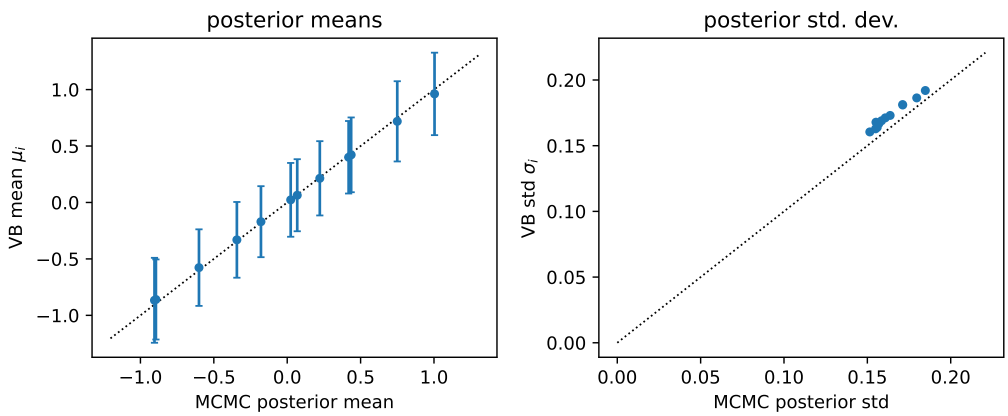

Figure 3: Variational posterior versus a reference Metropolis sampler on the exact logistic model ($n = 12$). Left: posterior means agree closely (error bars are the VB 95% intervals). Right: per-item posterior standard deviations of the centred scores — i.e. on the identified scale, with the unidentified global level removed — agree between VB and MCMC.

Active selection. We compare the three acquisition criteria of Section 7 against choosing pairs uniformly at random. Each run starts with no observations and adds one comparison at a time; after every comparison we refit the posterior and record the number of inversions of its ranking against the ground truth. We use $n = 12$ items, run for 1000 comparisons, and average over 800 fresh random instances. Figure 4 shows the result. Information gain and the sum of posterior variances are essentially indistinguishable and best, converging fastest and to the fewest inversions (final means 4.25 and

4.23, and averaged over the whole run 7.34 and 7.41). Uniform-random selection is a close but consistently worse baseline (4.64; 7.89). Strikingly, greedily minimising the *expected number of inversions* — the very quantity being measured — is the *worst* of the four (4.53; 8.86): as a one-step rule it is exploitative, repeatedly probing the current near-ties rather than spreading information, and it converges more slowly than random. The two uncertainty-directed criteria, which shrink posterior uncertainty broadly instead of chasing the current decision boundary, explore more efficiently and win. We do not claim a specific mechanism beyond this greedy myopia; a lookahead or tree search over the same gradient (Section 7) would likely temper it.

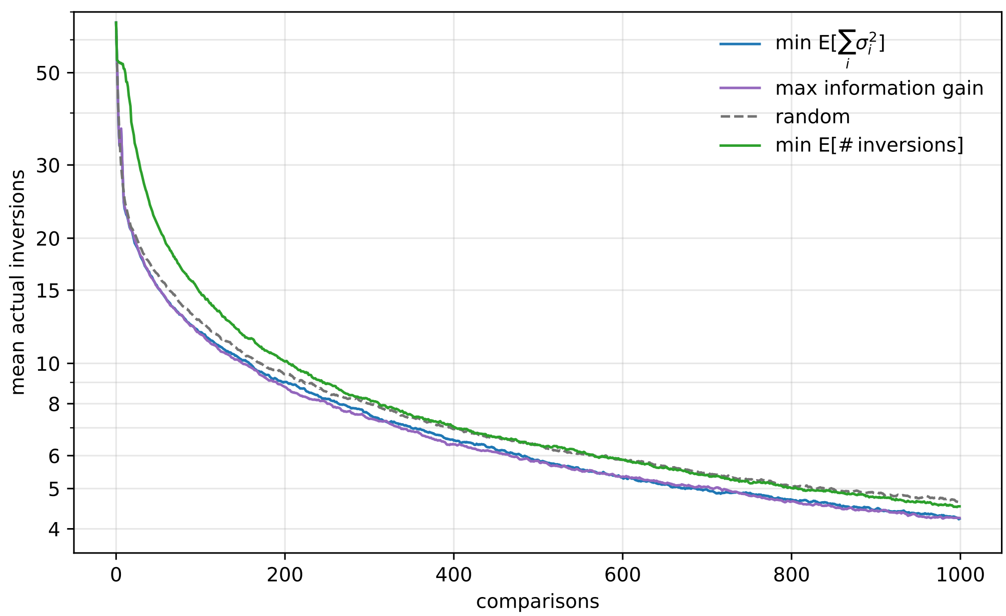

Figure 4: Active pair selection ($n = 12$, averaged over 800 fresh instances). Mean number of inversions of the posterior ranking against the ground truth versus the number of comparisons (logarithmic y-axis). Maximising information gain and minimising the sum of posterior variances are indistinguishable and best; greedily minimising the expected number of inversions is the worst, below even uniform-random selection.

Performance. To give a concrete sense of speed, on a single core of an AMD Ryzen 7 3700X our C++ implementation (compiled with `-O3`) fits a problem with $n = 80$ items in about 0.7 s, $n = 200$ items (around 1.6×10^5 comparisons) in about 8 s, and $n = 360$ items (around 5×10^5 comparisons) in about 33 s.

Verification. We checked the closed-form gradient and Hessian — including the correction terms of Section 5.3 — against central differences, matching to better than 10^{-8}, and likewise the active-selection tangent: the predicted marginal effect of an (infinitesimal) comparison on each loss matches a central-difference estimate across all pairs.

Conclusion

Approximating the logistic link with an error function turns hierarchical Bayesian ranking into a problem with a fully closed-form variational objective, gradient and Hessian. The closed form has no business working out as cleanly as it does, but it does, and it buys two things: inference fast enough to scale to many items or to support deep backtracking search, and a fitted posterior that is differentiable in the data, which we exploit for active next-pair selection. Our experiments confirm that the approximation closely matches an exact-model MCMC reference.

The model could be extended to richer outcomes — draws, more than two competitors per match, and scores that drift over time — much as TrueSkill [HMG07] handles these; whether the closed form survives these extensions is an open and tempting question.

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